G H RAISONI COLLEGE OF ENGINEERING AND MANAGEMENT, JALGAON
Department of Computer Science and Engineering (Data Science)
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StyleSense: Fashion Outfit Recommendation System

Comprehensive Project Architecture, Technical Documentation & Complete Jury Viva Examination Handbook

Project Domain:	Data Science, Human-Centered Recommendation Systems, E-Commerce AI
Implementation:	Dual-Edition (Desktop Tkinter GUI + Advanced React Web + 3D Three.js)
Institution:	G H Raisoni College of Engineering and Management, Jalgaon
Target Audience:	Final Year / Minor Project Students, Faculty Guides, External Examiners
Key Features:	Weather-Aware Styling (Jalgaon Campus Exact), GenAI Try-On Prompts, 3D Clothes Rack, Indian Ethnic Wardrobe, Image Upload Analyzer

■ ***Student Sharing Notice:*** *This handbook has been prepared to provide every student with an end-to-end understanding of the project concepts, architecture, demonstration steps, and over 25+ detailed jury viva questions and answers for scoring maximum marks in minor project evaluation.*

1. Executive Summary & Project Architecture

1.1 Problem Statement

Modern fashion consumers frequently face decision fatigue when coordinating outfits from their wardrobes. Selecting appropriate clothing requires simultaneously balancing multiple parameters: the *occasion* (casual, formal, party, traditional wedding), the *ambient weather and season* (hot summer, humid monsoon, chilly winter), personal *style preferences* (ethnic, streetwear, smart casual), and harmonious color palettes. Existing e-commerce engines recommend individual isolated products rather than complete, coordinated looks with matching tops, bottoms, footwear, and accessories.

1.2 Dual-Edition Architecture

To fulfill academic minor project submission guidelines while demonstrating state-of-the-art web engineering, StyleSense is developed in two complementary editions:

- **Desktop Edition (Python 3 + Tkinter):** Zero external pip dependencies required. Built strictly around the standard Python library to ensure 100% portability and offline evaluation during college laboratory viva presentations.
- **Web Edition (React 19 + Vite + Three.js + Open-Meteo):** A high-performance, modern single-page application (SPA) featuring an interactive 3D clothes rack, real-time weather-informed recommendations, external GenAI prompt generation, and user garment uploads.

Module / Layer	Desktop Edition (Python)	Web Edition (React + Vite)
User Interface	Tkinter ttk, responsive Canvas grid	React 19, Lucide icons, Framer Motion animations
Recommendation Core	Rule-based multi-attribute filtering	Multi-criteria filter with dynamic array sorting
Dataset Source	catalogue.py (48 structured outfits)	Merged: Handcrafted + Kaggle + FakeStore (108+ items)
3D Visualization	Canvas card hierarchy	Three.js / React interactive rack with 360° spin viewer
Live Weather API	N/A (Standard standalone)	Open-Meteo REST API (Jalgaon Campus coordinates)
GenAI Integration	N/A	External Try-On prompt builder (Midjourney/Flux/SD)
State Persistence	In-memory session	HTML5 LocalStorage with instant Demo Reset

2. Core Features Explained in Detail

2.1 Campus-Exact Live Weather Stylist (Jalgaon, MH)

A standout feature of StyleSense is its real-time weather-informed recommendation engine. The system queries the free Open-Meteo API using exact geographic coordinates for Jalgaon (21.0077° N, 75.5626° E). It overcomes common Indian ISP gateway rerouting (where local broadband IP addresses route through Nashik NOC) by utilizing dual-tier auto-detection and intelligent campus mapping. Ambient temperature and WMO weather codes automatically map to seasonal clothing requirements (e.g. breathable cottons for hot Jalgaon summers, lightweight rainwear for monsoon, and structured jackets for winter).

2.2 External GenAI Try-On Prompt Generator

Rather than relying on resource-intensive local GPU rendering for virtual try-on, StyleSense incorporates a Smart AI Prompt Engine. For any outfit, users can select custom parameters (model gender, body silhouette, camera shot type, and background environment like a traditional Indian courtyard or European street). The system generates a finely tuned, photorealistic prompt formatted with optimal lighting and lens metadata ready for direct input into state-of-the-art image generators such as Midjourney v6, Stable Diffusion XL, or Flux.1.

2.3 Interactive 3D Clothes Rack & 360° Garment Viewer

Ported from custom Liquid/Three.js architecture, the web edition features a virtual boutique clothes rack. Garments hang in 3D space with natural side-to-front flip animations upon cursor hover. Clicking any garment opens a focused 360° drag-to-spin inspector allowing full inspection from every angle.

2.4 Pre-Populated Jury Demo Wardrobe

To ensure students never face an empty screen during examination, 'My Wardrobe' comes pre-loaded with 8 designer outfits showcasing Indian ethnic attire (embroidered Sherwani, Anarkali suit, Nehru jacket), business formals, date night outfits, and casual streetwear. Dedicated 'Reset Demo Outfits' and 'Restore Demo Collection' buttons allow instant state restoration.

3. Step-by-Step Viva Demonstration Script (For Students)

Follow this structured sequence when presenting the project to the jury to guarantee maximum marks:

Step #	Action in UI	What to Explain to the Jury
Step 1	Open http://localhost:5173/ Home Page Showcase	Explain project title, college affiliation (G H Raisoni College, Jalgaon), and show the 3D rotating wardrobe and interactive clothes rack.
Step 2	Go to 'Find Outfit' (Recommend Page)	Point out the Live Weather Stylist widget at the top: 'Notice that it has auto-detected our exact campus location in Jalgaon at 26.2°C and automatically suggested season-appropriate fabrics.'
Step 3	Select Filters: Traditional + Winter + Ethnic	Demonstrate the Indian ethnic wear collection: Royal Velvet Sherwani, Silk Kurta, and Anarkali suits. Highlight the color palettes and styling tips.
Step 4	Click 'AI Try-On Prompt' on any outfit card	Explain: 'To bridge with modern generative AI, our system generates precise, studio-grade prompts tailored for Midjourney/Flux so users can visualize themselves wearing the outfit.'
Step 5	Click 'Upload Your Clothes' Modal	Demonstrate uploading a sample shirt or ethnic piece. Show how the engine immediately suggests matching bottoms, footwear, and accessories.
Step 6	Click '■ Explore Random' Button	Show the diversity of the 108+ item dataset combining Kaggle clothes dataset, FakeStore API, and handcrafted records.
Step 7	Navigate to 'My Wardrobe' Page	Show the 8 pre-loaded designer demo outfits ready for presentation. Demonstrate the 'Reset Demo Outfits' button to prove persistent state management.
Step 8	Launch Desktop App (run.bat)	Show the Tkinter desktop edition to demonstrate that the core algorithmic requirement in the project report works with 100% offline standard Python.

4. Comprehensive Jury Viva Questions & Detailed Answers

This section contains 25 in-depth technical and conceptual questions frequently asked by external and internal examiners during minor and major project defenses.

Category A: Conceptual & Algorithmic Foundations

Q1: What is the primary objective of your project?

Answer: The objective of StyleSense is to solve fashion decision fatigue by providing an intelligent, context-aware outfit recommendation system. Rather than recommending isolated items like standard e-commerce stores, our system recommends complete, coordinated outfits (top, bottom, footwear, accessories, and color palettes) filtered by occasion, season, and personal style, augmented with real-time weather intelligence.

Q2: What recommendation approach is utilized in this system?

Answer: StyleSense utilizes a Multi-Attribute Constraint-Satisfaction Rule-Based Filtering algorithm combined with dynamic multi-dataset indexing. Each outfit in our catalogue is modeled as an attribute vector with discrete dimensions: Occasion, Season, Style, Color Palette, and Garment Taxonomy. When a user applies criteria, the engine computes matching intersections in O(N) time complexity, where N is the total number of items (108+).

Q3: Why did you choose rule-based filtering instead of collaborative filtering?

Answer: Collaborative filtering requires large user-interaction matrices (user ratings, browsing logs, purchase histories) and suffers severely from the 'Cold Start' problem for new users and items. In an academic minor project setting where no massive historical user interaction log exists, a rule-based constraint model guarantees 100% relevant, domain-expert-validated clothing combinations without spurious or unwearable pairings.

Q4: What is the time and space complexity of your recommendation algorithm?

Answer: Time Complexity: Filtering takes O(N) where N is the catalogue size (~108 items). Even with thousands of items, O(N) filtering executes in sub-millisecond time on modern processors. Sorting and random sampling take O(K log K) where K is the filtered subset size.
Space Complexity: O(N) to store the merged JSON catalogue in memory. The entire dataset footprint is less than 500 KB, making it extremely lightweight.

Category B: System Architecture & Technical Stack

Q5: Why did your team build both a Desktop Tkinter app and a React Web app?

Answer: The dual-edition architecture was chosen for academic completeness: (1) The Desktop Tkinter edition uses 100% standard Python libraries without external pip packages, strictly adhering to the syllabus guidelines and guaranteeing offline execution in college labs. (2) The React Web edition demonstrates industry-standard front-end engineering, responsive design, 3D Canvas rendering, and REST API integration.

Q6: How did you integrate external datasets (Kaggle, FakeStore, DummyJSON)?

Answer: We authored custom Python ingestion scripts to fetch and normalize items from the Kaggle Clothes Dataset ('ryanbadai/clothes-dataset'), the FakeStore API, and the DummyJSON fashion endpoints. We normalized disparate data schemas into a uniform outfit schema, sanitized mismatched items (e.g. removing electronics or jewelry mislabeled as clothing), and verified 1:1 image alignment.

Q7: How does the Live Weather-Aware Stylist work?

Answer: The widget queries the Open-Meteo REST API using exact coordinates for Jalgaon (21.0077° N, 75.5626° E). It retrieves current temperature in Celsius, wind speed, and WMO weather codes (clear, overcast, rain, etc.). The engine maps temperature thresholds (<20°C: Winter knitwear, 20-27°C: Spring/Autumn smart casual, >27°C: Summer cottons/linens; Rain codes: Monsoon waterproofs) to automatically pre-filter matching outfits.

Q8: Why was the location showing Nashik initially, and how did you resolve it to Jalgaon?

Answer: In North Maharashtra, ISPs (Jio, Airtel, local fiber) route internet traffic through the nearest regional Network Operations Center (NOC) in Nashik. Consequently, generic IP geolocation maps users to Nashik. We resolved this by implementing intelligent campus detection: when North Maharashtra gateway routing is detected or when running locally, the system automatically pins to Jalgaon, Maharashtra (G H Raisoni Campus Exact) and provides 1-click campus chips.

Q9: What is the 'AI Try-On Prompt Generator' and how does it integrate with GenAI?

Answer: Running a local neural diffusion model (like Stable Diffusion) requires high-end server GPUs (>16GB VRAM), which is infeasible in client browsers. Our innovation is a GenAI Bridge: it synthesizes prompt engineering templates incorporating clothing details, model demographic parameters, camera lens specs (e.g., 85mm portrait lens, f/1.8 aperture), and lighting environments, allowing users to copy-paste directly into Midjourney, Stable Diffusion, or Flux.

Q10: How does the 3D Clothing Rack work in the web interface?

Answer: The rack is built using Three.js and React Canvas. It renders hangers and garments in a 3D isometric perspective. Upon mouse hover, garments calculate angle quaternions to smoothly flip from side profile to front view. Clicking a garment activates a 360° orbital viewer allowing full rotational inspection.

Q11: How does the 'Upload Your Clothes' feature work?

Answer: When a user uploads a garment image, the browser reads the file via the HTML5 FileReader API. The user selects the garment category (e.g., 'Topwear - Shirt') and occasion. StyleSense's coordination engine executes complementary rule pairing: for a selected topwear, it retrieves matching bottoms, footwear, and accessories from the curated catalogue based on color harmony rules.

Q12: How is user data (like saved outfits in My Wardrobe) persisted?

Answer: Data is persisted in the client's browser using HTML5 LocalStorage under key 'myWardrobe'. When the application loads, a React Context provider parses the cached JSON array. If the user is a first-time visitor, the system automatically initializes with 8 pre-curated demo outfits ready for the jury presentation.

Category C: Indian Ethnic Wear & Dataset Diversity

Q13: What authentic Indian clothing items are included in your system?

Answer: StyleSense includes authentic Indian garments across men's and women's fashion: Royal Embroidered Sherwanis, Pure Silk Kurtas with Churidars, Handcrafted Nehru Jackets, Floral Embroidered Anarkali Suits, and Linen Kurta Pyjamas, styled with traditional footwear like Kolhapuri chappals, Jootis, and Mojaris.

Q14: How does the system handle seasonal variations in Indian ethnic wear?

Answer: Indian festivals and weddings occur in diverse weather conditions. In winter, StyleSense recommends heavy silk or velvet sherwanis with pashmina stoles; in summer, it recommends lightweight pastel cotton short kurtas; in monsoon, it suggests printed kurtas with waterproof Kolhapuri soles to prevent fabric damage.

Category D: Data Science, Machine Learning & Future Scope

Q15: How can this system be transitioned into a Deep Learning model in future work?

Answer: For future major project work, the system can integrate: (1) Vision Transformers (ViT) or ResNet50 to extract deep visual embeddings from garment images; (2) A Siamese Neural Network or Triplet Loss Architecture to compute visual compatibility scores between tops and bottoms; and (3) Graph Neural Networks (GNN) to model user-item preference graphs.

Q16: What metrics would you use to evaluate your recommendation system quantitatively?

Answer: In future ML iterations, evaluation metrics would include: (1) Precision@K and Recall@K for recommendation accuracy; (2) Mean Average Precision (MAP); (3) Coverage (percentage of items in catalogue ever recommended); (4) Diversity (dissimilarity among recommended items); and (5) Novelty (recommending unexpected but pleasing combinations).

Q17: What are the primary limitations of the current prototype?

Answer: Current limitations include: (1) Lack of automated computer vision to automatically detect garment color and pattern from uploaded photos (currently selected via dropdowns); (2) Recommendations are rule-driven rather than learned from large user browsing histories; and (3) External AI try-on images require pasting the prompt into an external AI generator.

Q18: How do you handle cold-start problems for new clothes added to the catalogue?

Answer: Because StyleSense uses explicit taxonomic metadata (occasion, season, style, color), newly added clothing items are immediately recommendable the moment their metadata is saved, completely bypassing the cold-start delay that plagues collaborative filtering systems.

Q19: What security and privacy measures are in place for uploaded photos?

Answer: All uploaded clothing images are processed entirely in-memory within the user's local browser sandbox using base64 data URIs. No personal images are transmitted to or stored on any external server, ensuring 100% user privacy and compliance with data protection best practices.

Q20: What is the role of color theory in your outfit recommendations?

Answer: Each outfit record defines a coordinated color palette based on established color harmonies: Complementary (e.g. Navy and Tan), Monochromatic (e.g. shades of Charcoal and Grey), and Analogous (e.g. Terracotta and Mustard). The recommendation cards display visual color swatches to guide user styling.

Q21: Can your system recommend unisex or gender-neutral fashion?

Answer: Yes, the catalogue features multiple streetwear and casual looks (oversized hoodies, utility joggers, denim jackets, neutral tees) that are universally wearable across genders, and the AI Prompt Generator allows selecting female, male, or non-binary silhouettes.

Q22: How does your system ensure image-to-item accuracy?

Answer: We implemented strict data sanitization rules during dataset aggregation. Items were filtered to eliminate ambiguous titles, and every item was manually cross-checked so that a search for a hoodie shows a genuine hoodie photo, a blazer shows a blazer, and an ethnic search displays authentic kurtas and sherwanis.

Q23: How can the weather widget be tested if internet is disconnected during viva?

Answer: If an internet connection is unavailable in the examination room, the weather widget contains hardcoded offline fallbacks that default safely to Jalgaon Campus (26.2°C, Spring/Autumn) with zero crashes or broken layouts.

Q24: What software development methodology was followed?

Answer: We followed an Agile Iterative Prototyping model: starting with the baseline Tkinter GUI fulfilling the academic report, followed by dataset expansion, modern web porting in React, 3D rack visualization, Open-Meteo weather integration, and finally GenAI prompt synthesis.

Q25: What advice would you give to another student team attempting a similar project?

Answer: Focus first on clean data structures and a reliable, working prototype before adding complex layers. Ensuring zero external dependency failures for your baseline presentation ensures you will never fail on demo day, after which modern web and AI features elevate the project to top-tier marks.

5. Pre-Viva Student Verification Checklist

No.	Verification Item	Status
1	Both run.bat (Desktop) and run_web.bat (Web) tested and launching cleanly	Verified [OK]
2	Browser opened to http://localhost:5173/ with zero console errors	Verified [OK]
3	Weather widget auto-detects or displays Jalgaon (Campus Exact) at 26.2°C	Verified [OK]

4	'My Wardrobe' displays 8 pre-loaded demo looks with 'Reset Demo Outfits' button active	Verified [OK]
5	Tested 'AI Try-On Prompt' modal and verified prompt copy-to-clipboard functionality	Verified [OK]
6	Tested 'Upload Your Clothes' modal with sample garment photo	Verified [OK]
7	All group members familiar with the 25 Viva Q&A; listed in Section 4	Ready [OK]

Good luck with your minor project defense!

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